

Haneul Yoo

Incoming Assistant Professor/Faculty Fellow · NYU

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Research Interests

My research interests lie at the intersection of machine learning (ML) and natural language processing (NLP), driven by a goal of inclusive AI.

- **Multilingual NLP:** I develop models and methods to effectively and efficiently improve generalization across languages [C8, P3]
- **Resources & Evaluation:** I design resources and evaluation frameworks, especially for underrepresented languages and cultures [C9, C4, J1, P4, W7]
- **AI for Impactful Domains:** I apply AI in the high-stakes, real-world domains for social good [C10, C6, C2, W1]

Current Employment

New York University (NYU)

Assistant Professor/Faculty Fellow

NY, United States

Starting Sep 2026

- Department of Computer Science, NYU Courant Institute School of Mathematics, Computing, and Data Science
- Advisor: Kyunghyun Cho

Education

Korea Advanced Institute of Science and Technology (KAIST)

Ph.D. in School of Computing

Daejeon, Korea

Sep 2022–Aug 2026

- Advisor: Alice Oh
- Dissertation: Enhancing multilingual large language models with code-switching
- Committee: Edward Choi, Minjoon Seo, Kyunghyun Cho, Hwaran Lee

Korea Advanced Institute of Science and Technology (KAIST)

M.S. in School of Computing

Daejeon, Korea

Sep 2020–Aug 2022

- Advisor: Alice Oh
- Thesis: Language learners in NLP data annotation
- Committee: So-Yeon Ahn, Uichin Lee

Ajou University

B.S. in Computer Science and Engineering

B.S. in Digital Media

Suwon, Korea

Mar 2017–Aug 2020

- Summa Cum Laude, Rank: 1st in the department
- **University of Wisconsin, WI, USA** — *Semester Abroad Winter 2018*

Past Employment

Korea Advanced Institute of Science and Technology (KAIST)

Research Assistant

Daejeon, Korea

Sep 2020–Aug 2026

- Advisor: Alice Oh

New York University (NYU)

Visiting Scholar

NY, United States

Sep 2025–Feb 2026

- Development of robust, controllable, and unbiased trustworthy AI technology
- Host: Kyunghyun Cho

Boostcamp AI Tech

Lecturer

- Course: Data-Centric AI

Remote

Aug 2024–Feb 2025

NAVER AI Lab

Research Intern

- Development and evaluation of multilingual LLMs using code-switching corpora
- Host: Hwaran Lee

Seongnam, Korea

Mar 2024–Aug 2024

Upstage

Data Management Intern

- Construction and management of Korean evaluation benchmark for LLMs
- Hosts: Jiyeon Han and Taehwan Oh

Remote

Aug 2023–Feb 2024

NAVER AI Lab

Visiting Student

- Construction and management of KoBBQ, Korean bias benchmark for question answering
- Host: Hwaran Lee

Seongnam, Korea

Mar 2023–Jul 2023

Ajou University

Undergraduate Research Assistant

- Study of deep Learning model for bio-signal (EEG) analysis
- Advisor: Kyung-Ah Sohn and Jungryul Seo

Suwon, Korea

Mar 2020–Aug 2020

KEPCO Research Institute (KEPRI)

Research Intern

- Development of application services utilizing electric used power data
- Hosts: Moonsuk Choi, Inji Choi

Daejeon, Korea

Sep 2019–Feb 2020

Commonwealth Scientific and Industrial Research Organisation (CSIRO)

Research Intern

- Development of Convolutional deep neural network for detecting cattle in images/videos
- Host: Brano Kusy

QLD, Australia

Jul 2019–Aug 2019

Publications

* denotes equal contributions among authors

Google Scholar: <https://scholar.google.com/citations?user=9voXzU0AAAAJ>

JOURNAL ARTICLES

- [J1] Jiho Jin*, Jiseon Kim*, Nayeon Lee*, **Haneul Yoo***, Alice Oh, Hwaran Lee. “KoBBQ: Korean Bias Benchmark for Question Answering.” *TACL*. 2024. 🏆 **Oral at ACL**

CONFERENCE PROCEEDINGS

- [C13] Juhyun Oh, **Haneul Yoo***, Faiz Ghifari Haznitrana*, Alice Oh. “OLA: Output Language Alignment in Code-Switched LLM Interactions.” *ACL*. 2026.
- [C12] Seyoung Song, Nawon Kim, Songeun Chae, Kiwoong Park, Jiho Jin, **Haneul Yoo**, Kyunghyun Cho, Alice Oh. “Open Korean Historical Corpus: A Millennial-Scale Collection of Public Domain Texts.” *LREC*. 2026.
- [C11] Seyoung Song, **Haneul Yoo**, Jiho Jin, Kyunghyun Cho, Alice Oh. “Shared Heritage, Distinct Writing: Rethinking Resource Selection for East Asian Historical Documents.” *Findings of IJCNLP-AACL*. 2025.
- [C10] **Haneul Yoo**, Jieun Han, So-Yeon Ahn, Alice Oh. “DREs: Dataset for Rubric-based Essay Scoring on EFL Writing.” *ACL*. 2025.

- [C9] **Haneul Yoo**, Yongjin Yang, Hwaran Lee. “Code-Switching Red-Teaming: LLM Evaluation for Safety and Multilingual Understanding.” *ACL*. 2025. 🏆 Best Poster Award at Pre-ACL Workshop
- [C8] **Haneul Yoo**, Cheonbok Park, Sangdoo Yun, Alice Oh, Hwaran Lee. “Code-Switching Curriculum Learning for Multilingual Transfer in LLMs.” *Findings of ACL*. 2025.
- [C7] Yongjin Yang, **Haneul Yoo**, Hwaran Lee. “MAQA: Evaluating Uncertainty Quantification in LLMs Regarding Data Uncertainty.” *Findings of NAACL*. 2025.
- [C6] Jieun Han*, **Haneul Yoo***, Junho Myung, Minsun Kim, Tak Yeon Lee, So-Yeon Ahn, Alice Oh. “RECIPE4U: Student-ChatGPT Dialogue in EFL Writing Education.” *LREC-COLING*. 2024.
- [C5] Eunsu Kim, Juyoung Suk, Philhoon Oh, **Haneul Yoo**, James Throne, Alice Oh. “CLiCK: Evaluation of Cultural and Linguistic Intelligence in Korean.” *LREC-COLING*. 2024.
- [C4] **Haneul Yoo**, Rifki Afina Putri, Changyoon Lee, Youngin Lee, So-Yeon Ahn, Dongyeop Kang, Alice Oh. “Rethinking Annotation: Can Language Learners Contribute?.” *ACL*. 2023.
- [C3] Juhee Son*, Jiho Jin*, **Haneul Yoo**, JinYeong Bak, Kyunghyun Cho, Alice Oh. “Translating Hanja Historical Documents to Understandable Korean and English.” *Findings of EMNLP*. 2022.
- [C2] **Haneul Yoo**, Jiho Jin, Juhee Son, JinYeong Bak, Kyunghyun Cho, Alice Oh. “HUE: Pretrained Model and Dataset for Understanding Hanja Documents of Ancient Korea.” *Findings of NAACL*. 2022.
- [C1] Yohan Jo, **Haneul Yoo**, JinYeong Bak, Alice Oh, Chris Reed, Eduard Hovy. “Knowledge-Enhanced Evidence Retrieval for Counterargument Generation.” *Findings of EMNLP*. 2021.

IN REVIEW & PREPRINTS

- [P8] Siddhesh Pawar*, Sarah Masud*, **Haneul Yoo**, Alice Oh, Isabelle Augenstein. “Not What, But How: A Framework for Auditing LLM Responses across Positioning, Generalization, Anthromorphism, and Maxims.” *Arxiv*. 2026.
- [P7] Nahyun Lee, Dongkeun Yoon, Guijin Son, Geewook Kim, Dayoon Ko, Jeonghun Park, **Haneul Yoo**, Jaewon Cho, Junghun Park, Changyoon Lee, Kyochul Jang, Jaeyeon Kim, Eunsu Kim, Woojin Cho, Seungone Kim. “K-BrowseComp: A Web Browsing Agent Benchmark Grounded in Korean Contexts.” *Arxiv*. 2026.
- [P6] Jieun Han, Junyeong Park, Seohyun Park, **Haneul Yoo**, Hyoungwook Jin, Xing Xie, Evelyne Viegas, Sean Rintel, Miran Lee, Alice Oh, So-Yeon Ahn, Fangzhao Wu. “Learner Behaviour in Learning by Teaching with Imperfect LLM Peers.” *Submitted to Nature Human Behaviour*. 2026.
- [P5] Genta Indra Winata, David Anugraha, Patrick Amadeus Irawan, Anirban Das, **Haneul Yoo**, Paresh Dashore, Shreyas Kulkarni, Ruochen Zhang, Haruki Sakajo, Frederikus Hudi, Anaelia Ovalle, Syrielle Montariol, Félix Gaschi, Michael Anugraha, Rutuj Ravindra Puranik, Zawad Hayat Ahmed, Adril Putra Merin, Emmanuele Chersoni. “Can Large Language Models Understand, Reason About, and Generate Code-Switched Text?.” *Arxiv*. 2026.
- [P4] **Haneul Yoo**, Won Ik Cho, Geunhye Kim, Jiyeon Han. “From National Curricula to Cultural Awareness: Constructing Open-Ended Culture-Specific Question Answering Dataset.” *Arxiv*. 2026.
- [P3] **Haneul Yoo**, Jiho Jin, Kyunghyun Cho, Alice Oh. “Code-Switching In-Context Learning for Cross-Lingual Transfer of Large Language Models.” *Arxiv*. 2025.
- [P2] Seyoung Song, **Haneul Yoo**, Jiho Jin, Kyunghyun Cho, Alice Oh. “HERITAGE: An End-to-End Web Platform for Processing Korean Historical Documents in Hanja.” *Arxiv*. 2025.
- [P1] HyperCLOVA X Team. “HyperCLOVA X Technical Report.” *Arxiv*. 2024.

EXTENDED ABSTRACTS, WORKSHOPS, POSTERS

- [W7] Eunsu Kim*, **Haneul Yoo***, Guijin Son, Hitesh Patel, Amit Agarwal, Alice Oh. “[Benchmarks Are Not Atomic: Composition-Aware LLM Evaluation using BenchHub](#).” *ICML Workshop on Combining Theory and Benchmarks*. 2026.
- [W6] Minsun Kim, SeonGyeom Kim, Suyoun Lee, Yoosang Yoon, Junho Myung, **Haneul Yoo**, Hyunseung Lim, Jieun Han, Yoonsu Kim, So-Yeon Ahn, Juho Kim, Alice Oh, Hwajung Hong, Tak Yeon Lee. “[Student-ChatGPT Interaction Visible: Designing a Teacher Dashboard for EFL Writing Education](#).” *LAK (Poster)*. 2026.
- [W5] Jieun Han, Daniel Lee*, **Haneul Yoo***, Jinsung Yoon, Junyeong Park, So-Yeon Ahn and Alice Oh. “[One-Topic-Doesn’t-Fit-All: Transcreating Reading Comprehension Test for Personalized Learning](#).” *EMNLP Workshop on Tailoring AI: Exploring Active and Passive LLM Personalization (PALS)*. 2025. 🏆 **Best Paper Award at PALS**
- [W4] Sunwoo Kim, **Haneul Yoo**, Alice Oh. “[On the Effect of Uncertainty on Layer-wise Inference Dynamics](#).” *ICML Workshop on Actionable Interpretability*. 2025.
- [W3] Jieun Han, **Haneul Yoo**, Junho Myung, Minsun Kim, Hyunseung Lim, Yoonsu Kim, Tak Yeon Lee, Hwajung Hong, Juho Kim, So-Yeon Ahn, Alice Oh. “[LLM-as-a-tutor in EFL Writing Education: Focusing on Evaluation of Student-LLM Interaction](#).” *EMNLP Workshop on Customizable NLP (CustomNLP4U)*. 2024.
- [W2] Jungyoub Cha, Jieun Han, **Haneul Yoo**, Alice Oh. “[CHOP: Integrating ChatGPT into EFL Oral Presentation Practice](#).” *EDM Workshop on Leveraging Large Language Models for Next Generation Educational Technologies*. 2024.
- [W1] Jieun Han*, **Haneul Yoo***, Yoonsu Kim, Junho Myung, Minsun Kim, Hyunseung Lim, Juho Kim, Tak Yeon Lee, Hwajung Hong, So-Yeon Ahn, Alice Oh. “[RECIPE: How to Integrate ChatGPT into EFL Writing Education](#).” *ACM Learning at Scale (L@S) Work in Progress*. 2023.

Talks

Jun 2026	Invited Talk at SKKU (Host: Prof. JinYeong Bak)	SKKU
Apr 2026	Invited Talk at AI4Good Kick-off Workshop (Host: Prof. Nuri Kim)	KAIST
Mar 2026	Invited Talk at NAVER AI Lab (Host: Dr. Sangdoo Yun)	NAVER Cloud
Feb 2026	Guest Lecture at Multilingual NLP Course (Host: Prof. Marine Carpuat)	UMD
Dec 2025	Invited Talk at McGill NLP Reading Group (Host: Prof. David Ifeoluwa Adelani)	Mila
Nov 2025	Invited Talk at Humans and Machines Lab (Host: Prof. Manoel Horta Ribeiro)	Princeton
Oct 2025	Invited Talk at Defense AI Program of Ministry of National Defense (Host: Prof. Byungkyoo Kim)	SKKU
Jul 2025	Invited Talk at Global AI Frontier Lab	NYU
Nov 2024	Invited Talk at KAIST exploreCSR	KAIST
Jun 2024	Invited Talk at KAIST exploreCSR	KAIST
Jun 2024	Invited Talk at NAVER AI Lab	NAVER Cloud
May 2023	Guest Lecture at Data-Centric AI Course (Host: Dr. Jiyeon Han)	Boostcamp AI Tech
Dec 2022	Guest Lecture at Data-Centric AI Course (Host: Dr. Jiyeon Han)	Boostcamp AI Tech

Teaching Experience

Lecturer Boostcamp AI Tech (2024–2025)

- Data-centric AI (Fall 2024)

TA KAIST (2021–2023)

- Machine Learning for Natural Language Processing (Fall 2023)
- AI Ethics (Spring 2023)
- AI and Its Social Impact (Spring 2023)
- Machine Learning (Fall 2022, Spring 2021)
- School of Computing Colloquium (Fall 2021)

TA Boostcamp AI Tech (2022–2023)

- Data-centric AI (Spring 2023, Fall 2022)

Lecturer Volunteer lecturer for high school students (2019–2020)

- Python at Suwon Information Science High School (Spring 2020)
- Java at Maetan High School (Spring 2019)

Advising Experience

Mentor Mentored 15 undergraduate/Master's students and 1 high school student at KAIST (2021–Present)

- David Okunoye (Spring 2026–Present), BS student at University of Lagos.
- Sunwoo Kim (Winter 2024–Winter 2025), BS →MS student at KAIST, Co-authored [W4].
- Inhwa Song (Fall 2024–Spring 2025), BS →PhD student at Princeton.
- Jungyoub Cha (Winter 2023–Summer 2024), BS →MS student at SNU, Co-authored [W2].
- Seyoung Song (Winter 2022–Winter 2023), BS →MS →PhD student at KAIST, Co-authored [C12, C11, P2].
- Jeongin Bae (Fall 2022–Spring 2023), BS →ML Engineer at NAVER Cloud.
- Darae Lee, Gimin Kang, Nayoung Oh, Rahul Agrawal, Akotet Yeshaw Tesema, Hyeonbin Hwang, Braahmi Padmakumar (Spring 2023), BS.
- Eunsu Kim (Spring 2023), BS →MS student at KAIST→PhD student at CMU, Co-authored [C5, W7].
- Jaehyun Kim (Summer 2021–Fall 2021), BS →MS student at KAIST.
- Taebin Yang (Spring 2022–Winter 2022), Gifted high school student.

Professional Services

Social Chair NeurIPS (2026)

Organizer COLM Workshop on Generative AI for the World (2026)

WiML Symposium at ICML (2026)

Satellite Event of ICML 2026 at KAIST (2026)

COLM Workshop on Multilingual and Equitable Language Technologies (2025)

The 2nd Workshop on Women in AI/EE/CS at KAIST (2024)

Pre-ACL International NLP Workshop at KAIST (2024)

Conference Reviewer EMNLP (2022) ACL Rolling Review (2023, 2024, 2025, 2026)

NeurIPS (2021, 2022, 2023, 2024, 2025, 2026)

ICASSP (2026)

COLM (2026)

ACM FAccT (2026)

ACM L@S (2026)

Ethics Reviewer ACL Rolling Review (2025, 2026)

Workshop Reviewer ACL Student Research Workshop (2026)
NeurIPS Workshop on Red Teaming GenAI: What Can We Learn from Adversaries? (2024)
EMNLP Workshop on Customizable NLP (2024)
ACL Workshop on Machine Learning for Ancient Languages (2024)

Student Volunteer ACM FAccT (2022)
COLING (2022)

Awards and Honors

PALS Workshop at EMNLP 2025	Best Paper Award	Nov 2025
Singapore AI Safety Red Teaming Challenge	3rd Place (Team Category)	Nov 2024
KAIST	Graduate Student Outstanding Paper Award	Oct 2024
Pre-ACL International NLP Workshop at KAIST	Best Poster Award	Aug 2024
BMW Summer School	Young Scientists—Travel Grants & Registration Fee Waiver	Jul 2024
Google exploreCSR	AI/CS/EE Rising Stars 2024	Jun 2024
BrainLink 2023	Invited Participants—Travel Grants & Registration Fee Waiver	Dec 2023
KAIST	KAIST Support Scholarship	Sep 2020–Aug 2026
Ajou University	Graduation with Honors—Summa Cum Laude (1st place)	Aug 2020
Gyeonggi Volunteer Center	Outstanding Volunteer Award	Jun 2020
Hyundai Asan Foundation	Academic Excellence Scholarship (merit-based)	Mar 2017–Aug 2020

Extracurricular Activities

Organizer Social Event, Global AI Frontier Lab at NeurIPS (2025)
Social Event, KAIST at NeurIPS (2023)
Birds-of-a-Feather, Women in AI/EE/CS at KAIST at NeurIPS (2024)
Birds-of-a-Feather, Women in AI/EE/CS at KAIST at ACL (2024)

Head of Secretariat Ajou University College Student Council (2019)

Staff NUBI Ajou (International student support & cultural exchange) (2018–2020)

References

Prof. Alice Oh KAIST (alice.oh@kaist.edu)

Prof. Kyunghyun Cho New York University (kyunghyun.cho@nyu.edu)

Prof. Hwaran Lee Sogang University (hwaranlee@sogang.ac.kr)